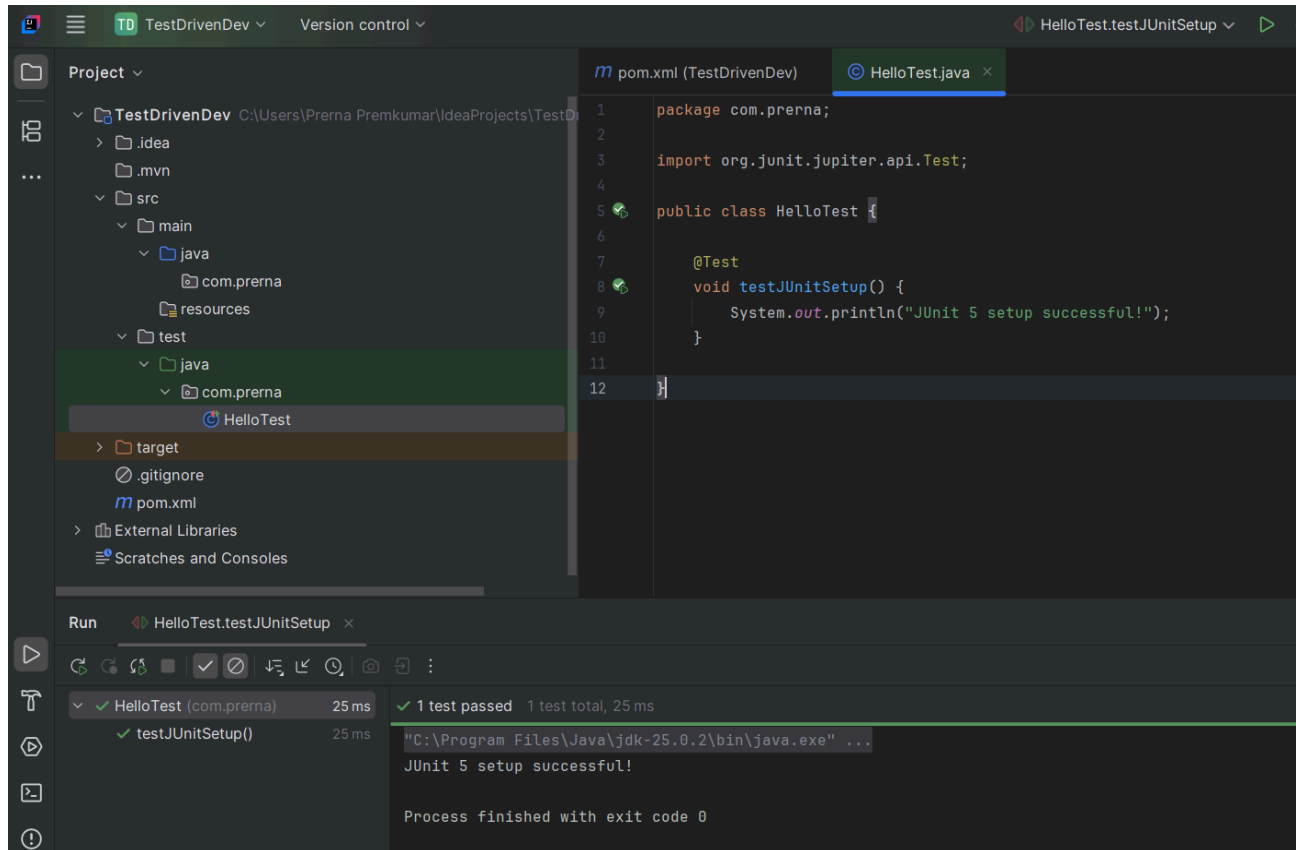


TEST DRIVEN DEVELOPMENT - JUNIT5

Exercise 1: Setting up JUnit5



Exercise 2:

Calculator.java:

```
package com.pruna.exercise2;

public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
    public int subtract(int a, int b) {
        return a - b;
    }
    public int multiply(int a, int b) {
        return a * b;
    }
    public int divide(int a, int b) {
        return a / b;
    }
}
```

CalculatorTest.java:

```
package com.pruna.exercise2;

import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

public class CalculatorTest {
    Calculator calculator = new Calculator();
    @Test
    void testAddition() {
        assertEquals(10, calculator.add(5, 5));
    }

    @Test
    void testSubtraction() {
        assertEquals(6, calculator.subtract(10, 4));
    }
}
```

```

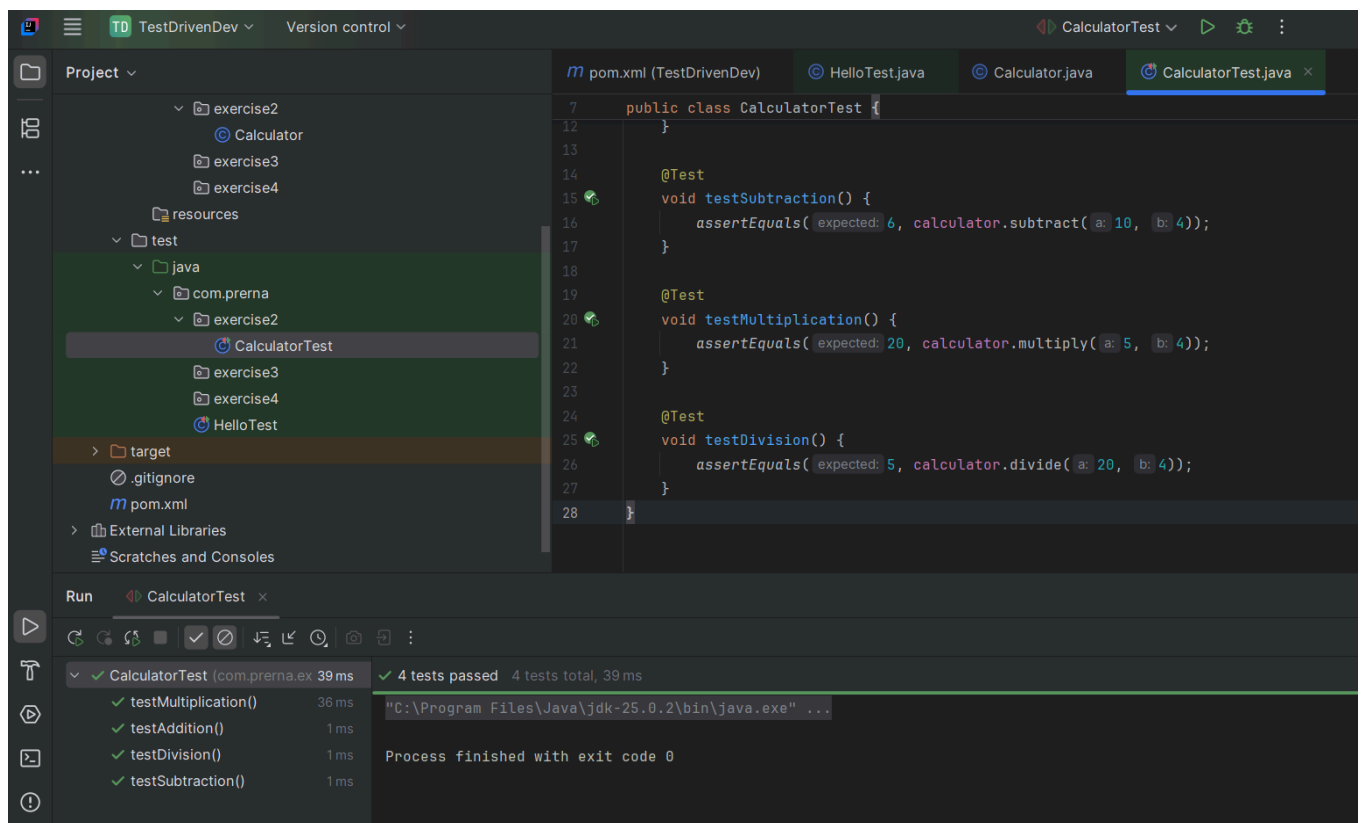
@Test
void testMultiplication() {
    assertEquals(20, calculator.multiply(5, 4));
}

```

```

@Test
void testDivision() {
    assertEquals(5, calculator.divide(20, 4));
}
}

```



Exercise 3: Assertions

AssertionsTest.java:

```
package com.perna.exercise3;

import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

public class AssertionsTest {
    @Test
    void testAssertions() {
        // Assert Equals
        assertEquals(5, 2 + 3);

        // Assert True
        assertTrue(5 > 3);

        // Assert False
        assertFalse(5 < 3);

        // Assert Null
        assertNull(null);

        // Assert Not Null
        assertNotNull(new Object());
    }
}
```

TestDrivenDev Version control

AssertionsTest

Project

exercise2

Calculator

exercise3

exercise4

resources

test

java

com.perna

exercise2

CalculatorTest

exercise3

AssertionsTest

exercise4

HelloTest

target

.gitignore

pom.xml

External Libraries

m pom.xml (TestDrivenDev)

HelloTest.java

Calculator.java

CalculatorTest.java

```
1 package com.perna.exercise3;
2
3 import org.junit.jupiter.api.Test;
4
5 import static org.junit.jupiter.api.Assertions.*;
6
7 public class AssertionsTest {
8     @Test
9     void testAssertions() {
10         // Assert Equals
11         assertEquals(expected: 5, actual: 2 + 3);
12
13         // Assert True
14         assertTrue(condition: 5 > 3);
15
16         // Assert False
17         assertFalse(condition: 5 < 3);
18
19         // Assert Null
20         assertNull(actual: null);
21     }
22 }
```

Run

AssertionsTest

AssertionsTest (com.perna.e) 49 ms

testAssertions() 49 ms

1 test passed 1 test total, 49 ms

"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" ...

Process finished with exit code 0

Exercise 4: Arrange-Act-Assert Pattern, Setup and Teardown

Calc.java:

```
package com.perna.exercise4;

public class Calc {
    public int add(int a, int b) {
        return a + b;
    }
}
```

CalcTest.java:

```
package com.perna.exercise4;

import org.junit.jupiter.api.AfterEach;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

public class CalcTest {
    Calc calculator;

    @BeforeEach
    void setUp() {
        calculator = new Calc();
        System.out.println("Setup completed");
    }

    @Test
    void testAddition() {
        int a = 10;
        int b = 20;
        int result = calculator.add(a, b);
        assertEquals(30, result);
    }

    @AfterEach
```

```

void tearDown() {
    System.out.println("Test completed");
}
}

```

